

BCCE 2026

What chemistry faculty believe about doctoral education

A phenomenographic and thematic characterization

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Supported by NSF CAREER #2142873 & #2602955 · Jordan Harshman, PI

Doctoral education trains the chemists who staff the discipline — and the enterprise is straining.

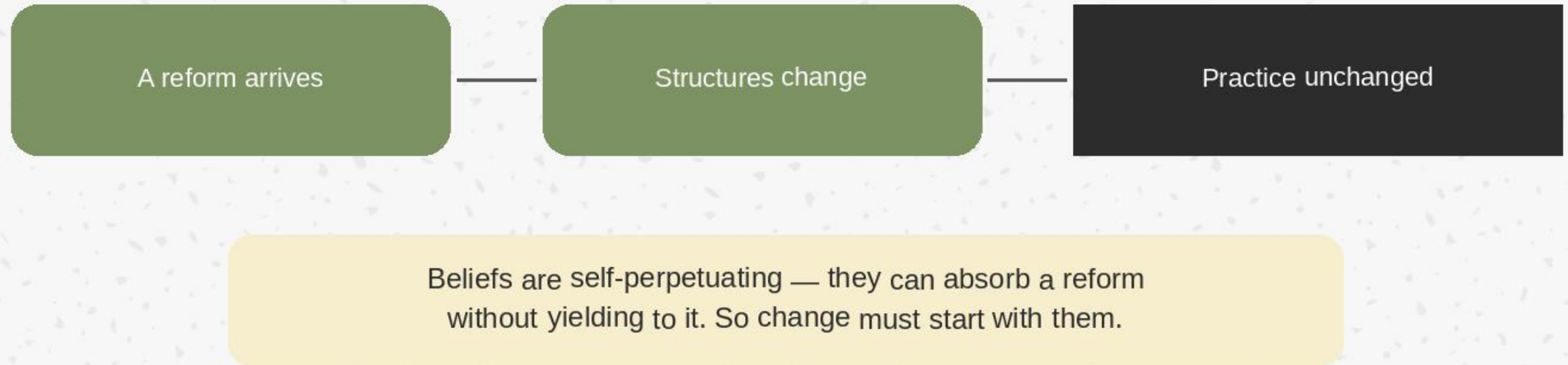
~2,900
chemistry PhDs each year

~196
U.S. doctoral programs

unchanged
the same structure for decades

Faculty name four strains: balancing competing responsibilities, no standard assessment of outcomes, uneven implementation, and inconsistent, largely untrained mentorship.

Reform changes the **structures** of the PhD, not what happens inside — because **beliefs absorb** it without yielding.



We know what faculty believe about teaching a class.
No one has characterized what they believe about advising a PhD.

TEACHING A CLASS

Beliefs well characterized —
and they predict practice

ADVISING A PhD

A different practice —
never characterized

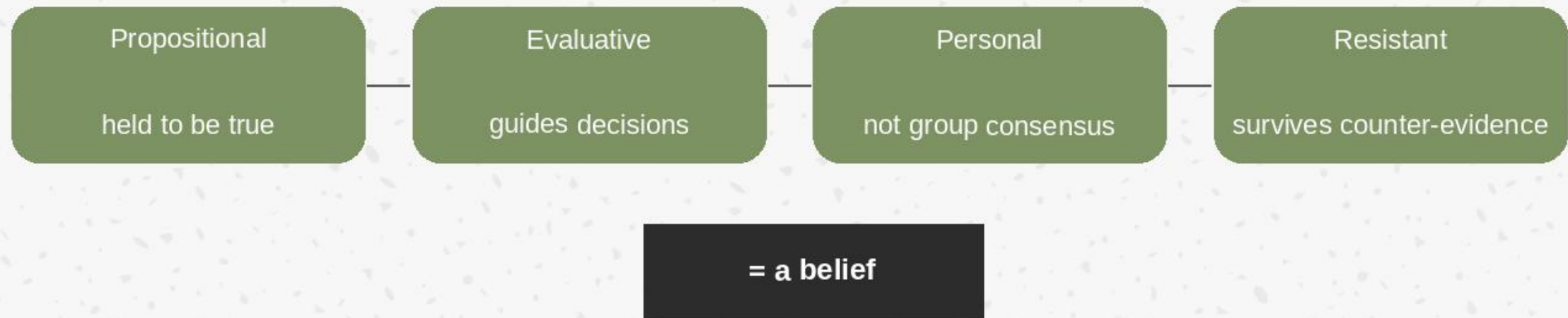
The closest prior study set beliefs aside — a proper look “*warranted its own study.*” **This is that study.**

RESEARCH QUESTION

What do chemistry faculty believe about doctoral education?

what it's for · how they learn · who develops which skill · the structure they work in

First, what counts as a **belief**? A proposition a person holds true and acts on — not a fact, and not easily changed.



We inferred beliefs from what faculty said and why — a knowledge claim answers to evidence; a belief is held on conviction.

We built those beliefs from twenty faculty's own words, coded to saturation.



No new codes appeared after interview 7 — the result held from 10 participants to all 20.

Here is the finding: faculty **agree** on the propositions —
and **divide** on exactly where each boundary falls.

AGREE
on four shared propositions

DIVIDE
on where the boundaries fall

“What differs is not which beliefs faculty hold, but where they draw the lines those beliefs require.”

The PhD builds a self-directed scientist —
in service of the student's own goals.

“When they start bringing ideas forward without necessarily being prompted.”

P2 — on what independence looks like

*“It's not my career.
It's not my goals.”*
— P2, on whose goals lead

WHERE THEY DIVIDE

answering existing questions

asking new questions

The realistic goal of the degree, versus the higher bar. Independence is the shared claim; its reach is the split.

Students learn by doing scaffolded research — but the same lonely start taught two faculty opposite lessons.

“Finding the answers on my own was not the most efficient — but it made me a much better scientist.”

P13 — credited the struggle

“Moving toward independence is really done in stages. Not something I tried to model at all.”

P7 — drew the opposite lesson

Both were left alone as students. The shared belief is learning-by-doing; the divide is how much unsupported struggle helps.

Development is a **shared, two-way job** — yet the same skill is one advisor's to teach and another's for the student to build.

"Creativity is the faculty's to teach."

— P5 · P17

"Creativity is the student's to build."

— P9 · P11

Every advisor draws a line between what they must teach and what the student must build — **and they draw it in different places.**

Structure shapes advising — and the **gaps** it leaves are the program's to fix, not the student's.

“Those professional skills often get kind of forgotten.”

P2 — on the program's gap

Fourteen of twenty
already name the gaps —
and can fix them.

WHERE THEY DIVIDE

the evolved system works

curricula & funding need change

On structure, many faculty are already critics rather than obstacles — they disagree on which gaps are real.

Within each shared theme, faculty differ along the axes below.
Each axis names the two poles of variation; nothing is plotted on it.



Agreement on every theme.
A **fault line** under every one.

Fifteen codes. Fifteen axes of division. Every faculty member lands somewhere on each — the agreement is real, and so is the disagreement one level down.

This gives reform its real map: shared ground to build on, and where “aligned” colleagues diverge.

SHARED PROPOSITIONS

Common ground — argue from where faculty already stand.

BOUNDARY DISAGREEMENTS

The real work — surface them, don't assume they're settled.

And convictions stay stable while practice is refined —
so support the implementation, not just the belief.

IN ONE LINE

Four shared convictions about the PhD —
and the lines faculty quietly draw differently.

Thank you.

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